BANKING CUSTOMER INTELLIGENCE

Exploratory Data Analysis Report

A documented review of customer demographics, financial behaviour, product usage, retention patterns, churn, correlations and historical customer value segments.

DATASET	SCOPE
10,000 customer records 14 original columns	EDA only No model training or predictive results
Notebook figures 26 saved chart outputs	Question basis Banking Customer Intelligence EDA Roadmap

Sources used

1. project.ipynb — source code, saved chart outputs and displayed calculation results.

Report scope

This report explains only the EDA work present in the supplied notebook before the Feature Engineering section. All 26 saved graph images are reproduced. Roadmap items without a corresponding completed calculation or chart are not presented as completed.

Interpretation convention: Exited = 0 means retained; Exited = 1 means churned. For grouped churn charts, the notebook calculates the mean of Exited and multiplies it by 100. This is the observed churn percentage within each group.

1. Dataset Overview and Data Quality

The roadmap asks for dataset size, feature types, missing-value checks and duplicate analysis. The notebook implements these checks through df.shape, df.info(), df.isnull().sum(), df.describe() and duplicate/customer-ID checks.

CHECK	NOTEBOOK RESULT	INTERPRETATION
Dataset size	10,000 rows × 14 columns	The analysis begins with 10,000 customer records and 14 original fields.
Data types	9 integer, 2 float, 3 text columns	The notebook reports 11 numeric-typed columns and 3 text columns. IDs and row number are included in that count.
Missing values	0 missing values in every column	The null-count output shows complete records for the 14 original columns.
Duplicate rows	0	No exact duplicate rows were found by df.duplicated().
Duplicate customer IDs	0	All 10,000 CustomerId values are unique in the notebook check.
Zero balance	3,617 customers (36.17%)	A substantial share of records has a zero account balance. The notebook retains these as observed values.

Descriptive statistics used in the notebook

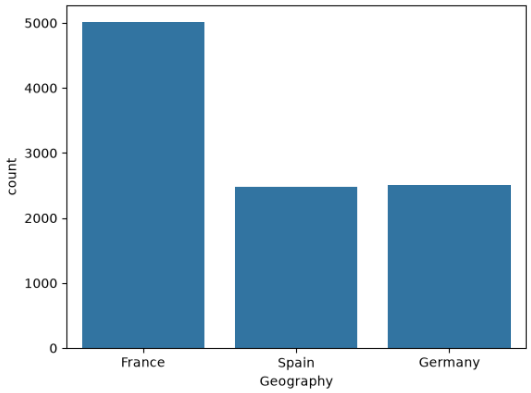
FEATURE	MEAN	MEDIAN	MIN–MAX
Age	38.92	37	18–92
CreditScore	650.53	652	350–850
Tenure	5.01	5	0–10
Balance	76,485.89	97,198.54	0–250,898.09
EstimatedSalary	100,090.24	100,193.92	11.58–199,992.48

Coverage note: The roadmap also proposes a feature-type distribution chart and a missing-value heatmap. Those visualizations are not present in the notebook; the underlying type and null checks are reported here from the executed outputs.

2. Customer Demographic Analysis — Geography and Gender

The roadmap asks how customers are distributed by country and gender, and whether observed churn rates differ across these groups. The notebook uses count plots for group sizes and grouped means of Exited for churn percentages.

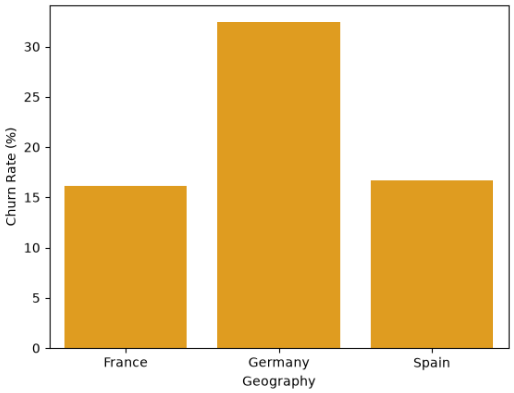
Customer Distribution by Geography



Question: How many customers are recorded in France, Germany and Spain?

Interpretation: France has 5,014 customers, Germany 2,509 and Spain 2,477. France represents just over half of the dataset; Germany and Spain have similar sample sizes.

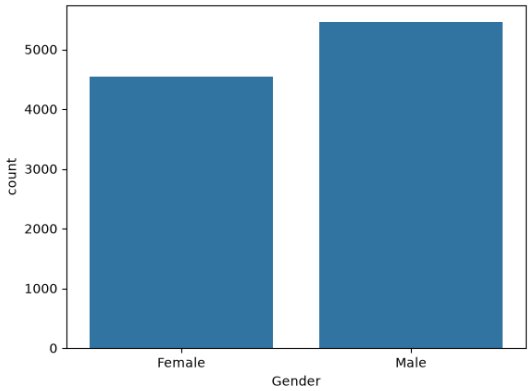
Churn Rate by Geography



Question: Does the observed churn percentage differ by country?

Interpretation: Germany has the highest observed churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain. The German rate is roughly twice the rates in the other two countries. This is an association, not proof that geography causes churn.

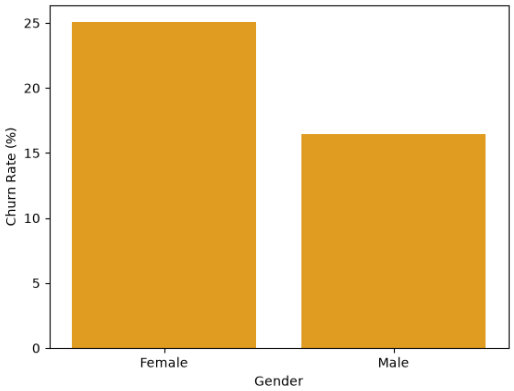
Customer Distribution by Gender



Question: How are customers distributed between male and female groups?

Interpretation: The dataset contains 5,457 male customers and 4,543 female customers. The sample has somewhat more male than female customers.

Churn Rate by Gender



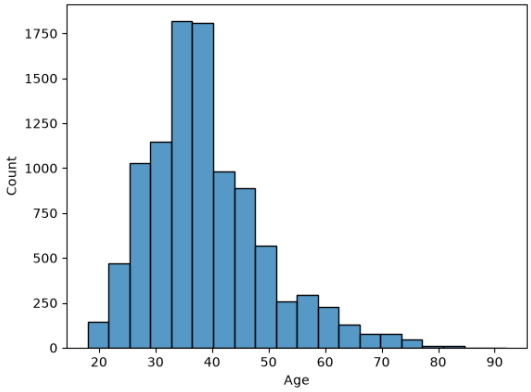
Question: Is the observed churn percentage different between genders?

Interpretation: The saved chart shows a higher churn percentage for female customers than for male customers. Exact group percentages are not printed in the saved notebook output, so this report does not assign precise values to the bars.

3. Customer Demographic Analysis — Age

The age analysis covers the overall age distribution, age differences between retained and churned customers, churn by exact age and churn by the age bands created in the notebook.

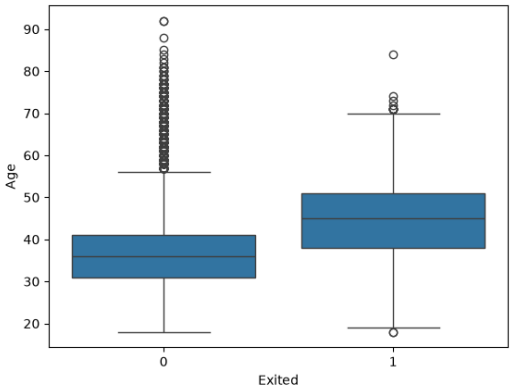
Age Distribution



Question: What does the customer age distribution look like?

Interpretation: Customer age ranges from 18 to 92 years. The mean age is 38.92 and the median is 37. Most observations are concentrated around the late twenties to early forties, with fewer customers at older ages.

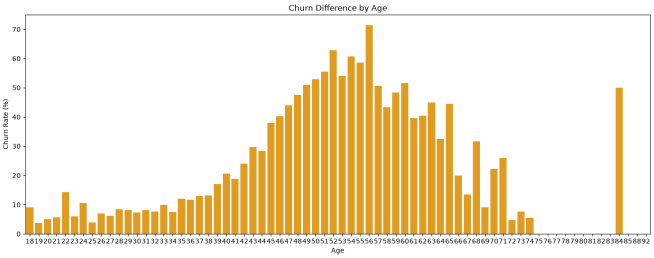
Age: Retained vs Churned



Question: Do the age distributions differ between customers who stayed and those who exited?

Interpretation: The churned group has a visibly higher median age than the retained group. The boxplot also shows older-age observations in both groups. The chart describes group differences; it does not establish age as a cause of churn.

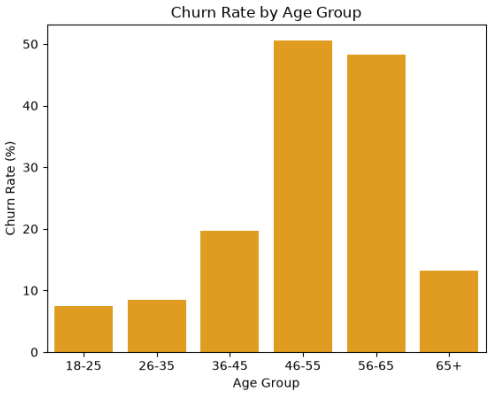
Churn Rate by Exact Age



Question: How does the observed churn rate vary across individual ages?

Interpretation: The rate varies across exact ages and is higher for many older age values. Individual-age groups can contain very different numbers of customers, so unusually high bars at particular ages should be read alongside each age group's sample size.

Churn Rate by Age Group



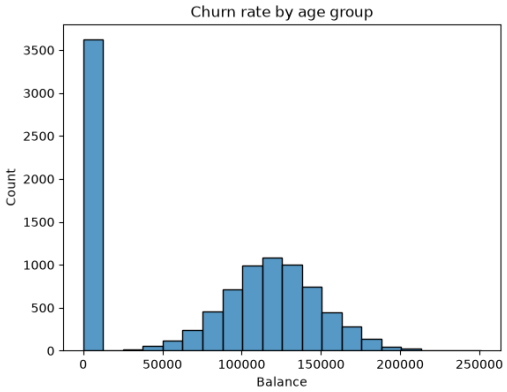
Question: Which of the notebook's age bands has the highest observed churn rate?

Interpretation: The 46–55 group has the highest rate (50.57%), followed by 56–65 (48.32%). Rates are 19.62% for 36–45, 8.50% for 26–35, 7.53% for 18–25 and 13.26% for 65+.

4. Financial Behaviour Analysis — Balance and Salary

The roadmap asks for distributions of account balance and estimated salary, plus comparisons between retained and churned customers. The notebook provides histograms and target-group boxplots.

Account Balance Distribution

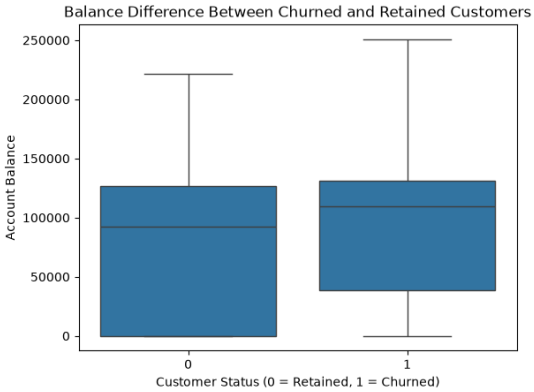


Question: How are customer account balances distributed?

Interpretation: The histogram contains a large spike at zero and a broad spread of positive balances. The notebook separately counts 3,617 zero-balance customers (36.17%). The mean balance is 76,485.89 and the median is 97,198.54.

Note: The saved plot title is outdated and refers to age-group churn. Its plotted variable is Balance; this report uses the correct description.

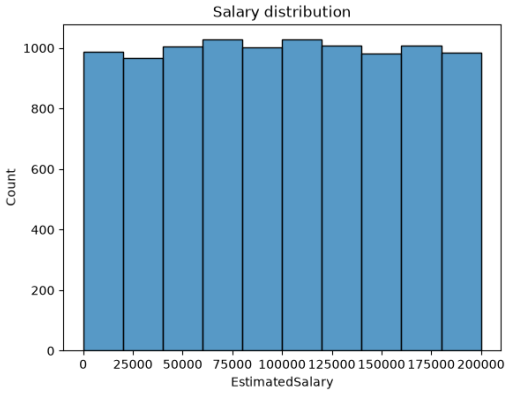
Balance: Retained vs Churned



Question: Do account balances differ between retained and churned customers?

Interpretation: The churned group has a higher mean balance (91,108.54) than the retained group (72,745.30). The boxplots show substantial overlap, so balance alone does not cleanly separate the two groups.

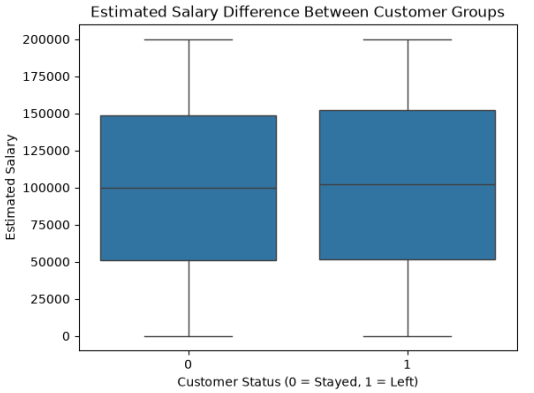
Estimated Salary Distribution



Question: How are estimated salaries spread across customers?

Interpretation: Estimated salaries extend from 11.58 to 199,992.48. The histogram is spread broadly across this range rather than concentrated in one narrow interval. The mean is 100,090.24.

Salary: Retained vs Churned



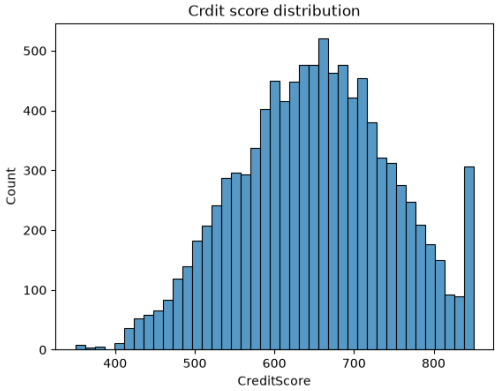
Question: Are estimated salaries visibly different between retained and churned customers?

Interpretation: The two salary distributions overlap heavily. The mean is 99,738.39 for retained customers and 101,465.68 for churned customers, a relatively small difference compared with the overall salary range.

5. Financial Behaviour Analysis — Credit Score

The notebook examines the credit-score distribution, groups scores into five categories, counts customers in each category and calculates observed churn percentages by category.

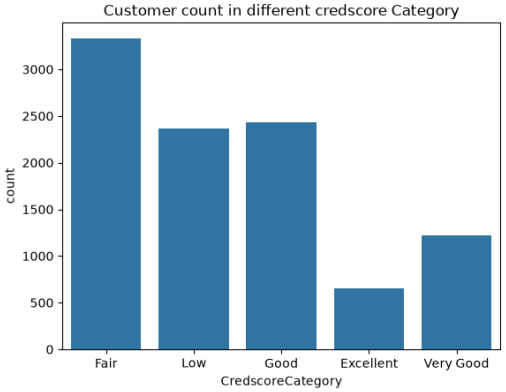
Credit Score Distribution



Question: How are customer credit scores distributed?

Interpretation: Credit scores range from 350 to 850. The mean is 650.53 and the median is 652. Most scores are concentrated around the middle of the scale.

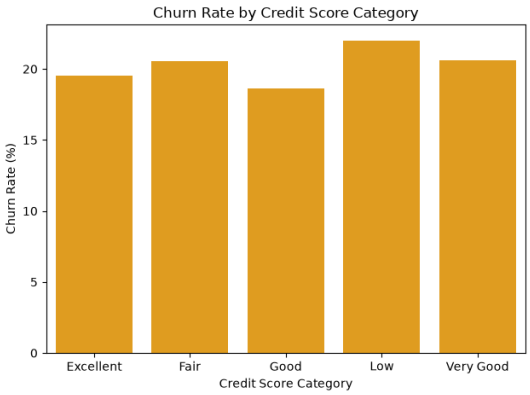
Customer Count by Credit Score Category



Question: How many customers fall into each credit-score band?

Interpretation: The notebook defines Low (<580), Fair (580–669), Good (670–739), Very Good (740–799) and Excellent (800 or above). Fair is the largest category in the saved count chart; Excellent is the smallest.

Churn Rate by Credit Score Category



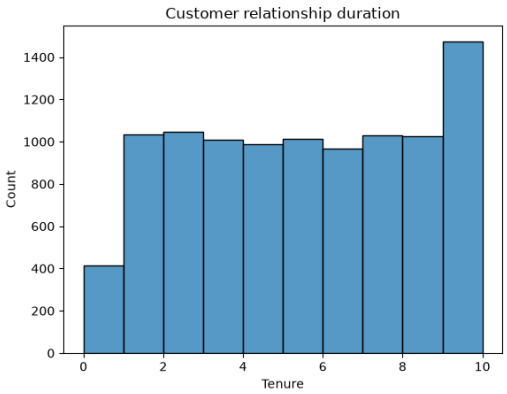
Question: Does observed churn vary across the five credit-score bands?

Interpretation: The five bars are relatively close, with churn percentages around the high-teens to low-twenties. Compared with the larger differences seen by geography or age group, credit-score category shows a smaller separation in this chart.

6. Customer Relationship and Product Usage

This section follows the roadmap's tenure and number-of-products questions: relationship duration, product ownership and the observed churn percentage within each group.

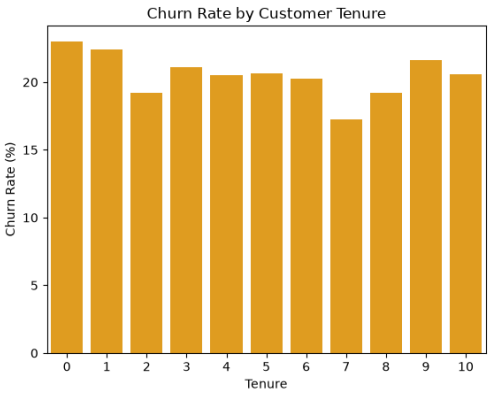
Customer Relationship Duration



Question: How long have customers maintained their relationship with the bank?

Interpretation: Tenure ranges from 0 to 10 years, with a mean of 5.01 years. Tenure values from 1 to 9 each have roughly one thousand customers, while 0 and 10 years have fewer.

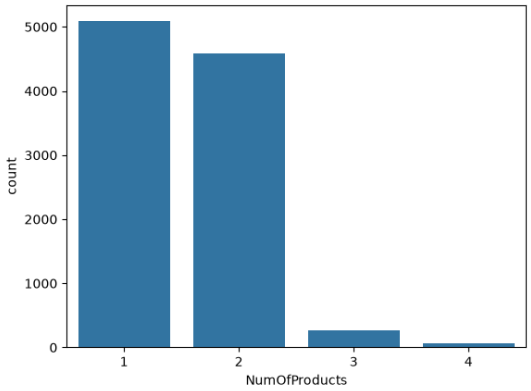
Churn Rate by Tenure



Question: Does observed churn change consistently with tenure?

Interpretation: The bars vary across tenure values but do not show a clear steadily increasing or decreasing pattern. Tenure alone therefore does not display a simple monotonic relationship with churn in this chart.

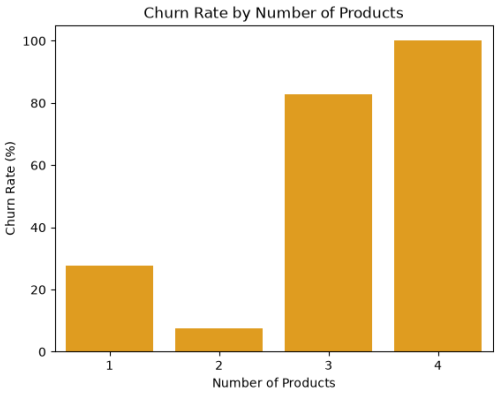
Customer Count by Number of Products



Question: How many products do customers hold?

Interpretation: Most customers hold one or two products: 5,084 have one product and 4,590 have two. Much smaller groups hold three products (266) or four products (60).

Churn Rate by Number of Products



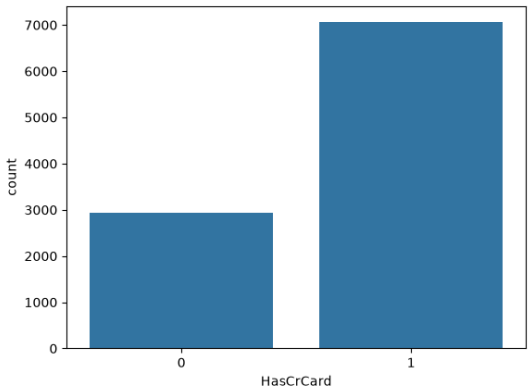
Question: How does the observed churn percentage differ by product count?

Interpretation: The chart shows higher churn rates among customers with three or four products than among those with one or two. However, the three- and four-product groups are small (266 and 60 customers), so their rates should be interpreted cautiously.

7. Product Usage and Customer Engagement

The notebook compares credit-card holders with non-holders and active members with inactive members. It also calculates retention by activity status.

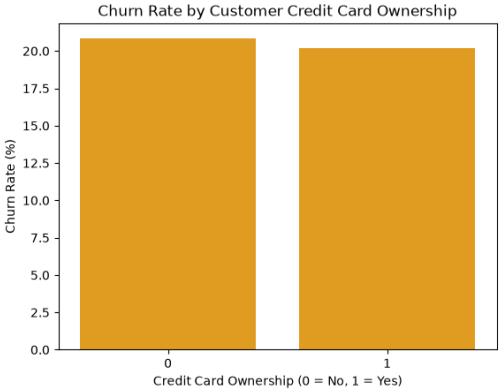
Customer Distribution by Credit Card Ownership



Question: How many customers have a bank credit card?

Interpretation: There are 7,055 credit-card holders and 2,945 customers without a credit card. Approximately seven in ten customers in the dataset have a card.

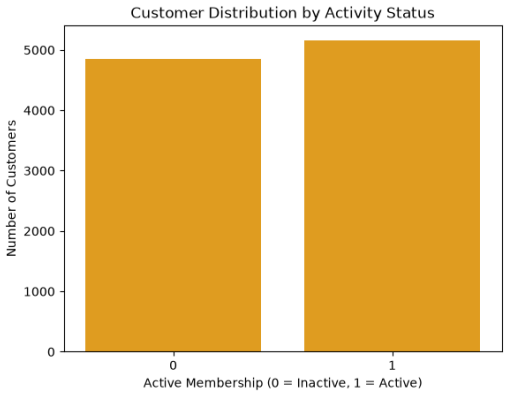
Churn Rate by Credit Card Ownership



Question: Does credit-card ownership correspond to a different observed churn rate?

Interpretation: The two bars are very close, around 20% churn. Credit-card ownership shows little separation in observed churn in this dataset and is not a strong standalone discriminator in this analysis.

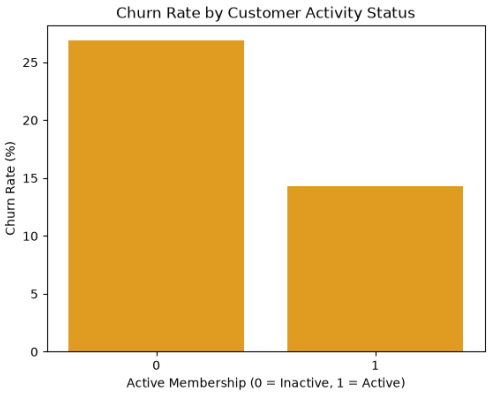
Customer Distribution by Activity Status



Question: How many customers are active and inactive members?

Interpretation: The dataset contains 5,151 active members and 4,849 inactive members. The two activity groups are similar in size.

Churn Rate by Customer Activity Status



Question: Is observed churn different for active and inactive members?

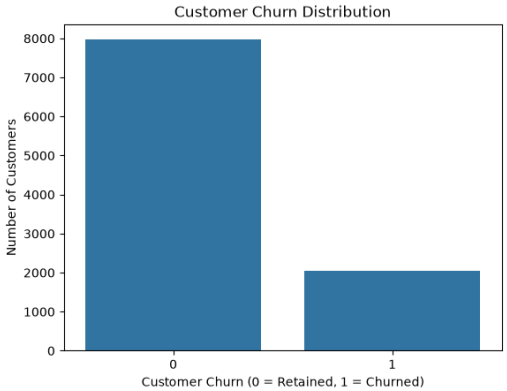
Interpretation: Inactive members have a churn rate of 26.85%, while active members have a rate of 14.27%. The inactive group therefore has a substantially higher observed churn percentage.

Retention result from the notebook: 73.15% of inactive members are retained, compared with 85.73% of active members. This is the complement of the churn rate because the target is binary.

8. Target Variable and Feature Relationship Analysis

The roadmap asks for the distribution of the target variable and a correlation analysis across selected numerical features. Both are present in the notebook.

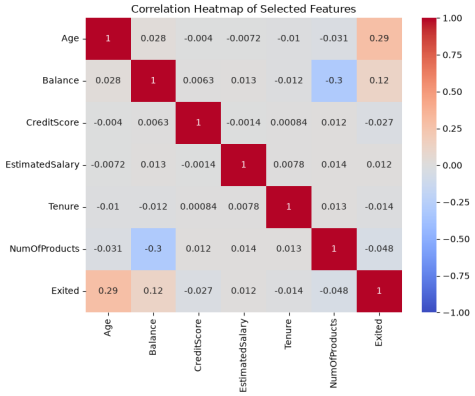
Customer Churn Distribution



Question: How many customers were retained and how many exited?

Interpretation: There are 7,963 retained customers (79.63%) and 2,037 churned customers (20.37%). The target is imbalanced: retained records are much more common than churned records.

Correlation Heatmap of Selected Features



Question: How are Age, Balance, CreditScore, EstimatedSalary, Tenure, NumOfProducts and Exited linearly associated?

Interpretation: Among the selected features, Age has the strongest positive correlation with Exited (0.29), followed by Balance (0.12). CreditScore (-0.027), EstimatedSalary (0.012), Tenure (-0.014) and NumOfProducts (-0.048) have correlations close to zero with Exited. Balance and NumOfProducts have a negative correlation of about -0.30.

Note: Correlation measures linear association, not causation. A weak correlation does not rule out non-linear or group-specific patterns.

9. Churn Customer Profile and Customer Value Matrix

The notebook compares mean customer characteristics by Exited status and then creates four historical groups using the median Balance and observed churn outcome.

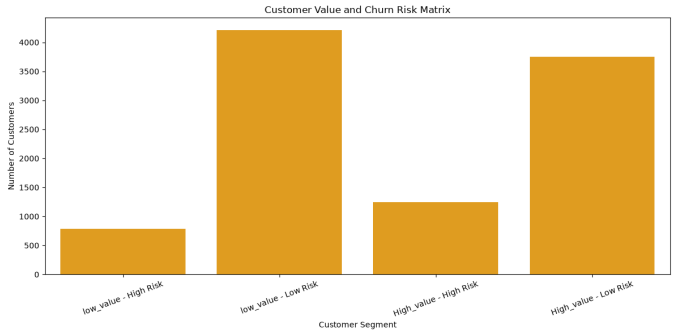
9.1 Retained vs churned customer profile

The means below reproduce the notebook's groupby('Exited').mean() table. The clearest differences are in age and balance; the other group means are relatively close.

FEATURE	RETAINED	CHURNED	OBSERVED DIFFERENCE
Age	37.41	44.84	Churned group is older on average.
Balance	72,745.30	91,108.54	Churned group has higher mean balance.
Estimated Salary	99,738.39	101,465.68	Means are relatively close.
Credit Score	651.85	645.35	Retained group is slightly higher.
Number of Products	1.54	1.48	Means are close.
Tenure	5.03	4.93	Means are close.

9.2 Customer value and historical churn-risk matrix

The notebook uses the median Balance of 97,198.54 as its value threshold. Balance at or above the median is High value; balance below the median is Low value. RiskStatus is mapped directly from Exited: retained = Low Risk; already churned = High Risk.



CUSTOMER SEGMENT	CUSTOMERS	SHARE
Low value – Low risk	4,212	42.12%
High value – Low risk	3,751	37.51%
High value – High risk	1,249	12.49%
Low value – High risk	788	7.88%

Limitation: this matrix uses Balance alone as the value proxy and the already-observed churn outcome as risk status. It is retrospective descriptive segmentation, not a prediction of future churn. The roadmap mentions other possible value indicators, but they are not used in this four-group matrix in the notebook.

SUPPLEMENTARY CUSTOMER SEGMENTATION ANALYSIS

K-Means Customer Segmentation and Behaviour Profiles

After the EDA and feature-engineering work, K-Means grouped the 10,000 customers into three segments using Age, Balance, NumOfProducts, IsActiveMember and HasZeroBalance. The cluster labels below describe the observed customer behaviour in each group.

Cluster profile comparison

PROFILE METRIC	CLUSTER 0	CLUSTER 1	CLUSTER 2
Business name	Active Balance-Holding	Inactive Balance-Holding	Predominantly Zero-Balance
Customers	3,278	3,104	3,618
Share of dataset	32.78%	31.04%	36.18%
Average age	39.84	38.52	38.44
Average balance	119,229.50	120,482.51	12.95
Median balance	119,624.57	120,135.42	0.00
Average products	1.38	1.39	1.79
Active member rate	100.00%	0.00%	51.77%
Zero-balance rate	0.00%	0.00%	99.97%
Historical churn rate	16.93%	31.60%	13.85%

What each customer segment represents

CLUSTER 0 · ACTIVE BALANCE-HOLDING CUSTOMERS

3,278 customers (32.78%)
All customers in this cluster are active members. Their average balance is 119,229.50 and they hold 1.38 products on average. The historical churn rate is 16.93%.

Behaviour: Customers with positive balances and active membership. This cluster has a lower churn rate than Cluster 1.

CLUSTER 1 · INACTIVE BALANCE-HOLDING CUSTOMERS

3,104 customers (31.04%)
No customers in this cluster are marked as active members. The average balance is 120,482.51 and average product count is 1.39. This cluster has the highest historical churn rate at 31.60%.

Behaviour: Customers with substantial balances but inactive membership. This is the segment with the highest observed churn.

CLUSTER 2 · PREDOMINANTLY ZERO-BALANCE CUSTOMERS

3,618 customers (36.18%)
The zero-balance rate is 99.97%. Average balance is 12.95 and median balance is zero. This cluster has the highest average product count (1.79), an active-member rate of 51.77%, and the lowest historical churn rate (13.85%).

Behaviour: Almost all customers have zero recorded balance, although they may hold multiple banking products.

Key observations

• **Cluster 1 has the highest historical churn rate (31.60%).** Its customers are inactive despite having an average balance similar to Cluster 0.

• **Cluster 2 is the largest segment (36.18%).** Nearly all of its customers have zero balance, yet this segment has the highest average number of products and the lowest historical churn rate.

• **Average age is similar across all three clusters** (approximately 38–40 years). In these profiles, account balance, activity status and product count distinguish the groups more clearly.

Interpretation note: *Exited was not used to form the K-Means clusters. The churn rate is calculated after clustering to describe the historical outcome within each segment; it is not an individual prediction of future churn. Cluster numbers are arbitrary model labels, so the business names describe the measured profiles rather than an inherent ranking.*

10. Roadmap Coverage

This table distinguishes roadmap questions supported by completed notebook work from those that are not yet represented by a completed calculation or saved graph.

ROADMAP AREA	STATUS IN NOTEBOOK	WHAT IS PRESENT
Dataset overview, missing values, duplicates	Completed checks	Shape, info, null counts, describe, duplicate rows and CustomerId checks.
Geography, gender and age	Completed	Counts, grouped churn rates, age histogram, age boxplot and age-group analysis.
Balance, salary and credit score	Partly completed	Balance/salary histograms and target-group boxplots; credit-score histogram, categories and grouped churn. No credit-score-by-target boxplot is present.
Tenure and product usage	Completed	Tenure distribution and churn; product counts and churn by number of products.
Credit-card and active membership	Completed	Group counts, churn rates and retention by activity status.
Target distribution and correlation	Completed	Exited count chart and selected-feature correlation heatmap.
Churn customer profile	Completed	Mean comparison for age, balance, salary, credit score, product count and tenure.
Customer value/risk matrix	Completed with a simple proxy	Four groups using median Balance and observed Exited outcome.
Feature-type chart; missingness heatmap/patterns	Not present	The notebook checks types and missing counts, but does not create the requested charts/pattern analysis.
Separate probability analysis; outlier analysis	Not separately completed	Grouped churn rates already calculate observed group churn proportions. A dedicated outlier investigation is not included in this notebook section.
Clustering, PCA, cluster profiles and feature interactions	Not included in this EDA section	No K-Means/PCA cluster output, cluster profile radar chart or age-balance/product-activity interaction chart is present before Feature Engineering.

How to read the coverage: grouped churn-rate calculations are already present for age, geography, gender, credit-score category, tenure, product count, credit-card ownership and activity status. Because each rate is calculated as the mean of the binary Exited column within a group, it is the empirical churn proportion for that group. A separate predictive probability model is not part of this EDA report.

11. Main Findings and EDA Conclusion

The following points summarize the strongest patterns directly supported by the notebook's executed calculations and saved graphs. They are descriptive findings from this dataset, not causal claims.

Geography: Germany has a substantially higher observed churn rate (32.44%) than France (16.15%) and Spain (16.67%).

Age: The 46–55 and 56–65 age groups have the highest observed churn rates in the notebook's age-band analysis.

Activity: Inactive members have a higher observed churn rate (26.85%) than active members (14.27%).

Product count: Customers with three or four products show high churn bars, but those groups are much smaller than the one- and two-product groups.

Credit card and credit score: Credit-card ownership and credit-score category show relatively small churn differences in the saved charts.

Correlation: Age has the strongest positive correlation with Exited among the selected numerical features (0.29); the other selected correlations with Exited are weak.

Customer profile: The churned group has higher mean age and balance than the retained group in the notebook's profile table.

Customer value/risk: The four-group matrix is based on median Balance and historical Exited status. It is useful as a descriptive summary but should not be interpreted as a future-risk model.

Conclusion

The completed EDA identifies differences in historical churn across geography, age, product count and activity status. It also shows substantial overlap in some financial distributions between retained and churned customers. These findings provide the descriptive basis for the next stages of the project: feature engineering, supervised churn modelling and customer segmentation. Those later stages are outside the scope of this report.

Source note: numeric findings in this report are taken from saved notebook outputs. Where the notebook saved a chart but did not preserve the underlying grouped percentages as text, the explanation describes the visible comparison without inventing exact values.